

Minimax Dual Certificates for the Common-Action Obstruction

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Abstract

The common-action obstruction certifies that no single instrument serves every branch of an information set; its polyhedral form carries a Farkas multiplier (Abaee, 2026b, Section 3.2). This paper states and proves the measure dual of the same obstruction: for control-affine dynamics with a compact convex control set, the common safe-action set is empty precisely when an adversarial probability measure on the active boundary–disturbance bundle drives the expected inward drift strictly negative against every control, with a certificate supported on at most $k+1$ points (k the control dimension; the bound is tight). The polyhedral case recovers the l_1 -normalized Farkas pair exactly; the singleton case recovers the Isaacs drift with the adversarial *distribution* over disturbances load-bearing. A two-action instance exhibits a strict minimax gap: without convexity the equivalence fails and no measure certifies — the audited finite codex systems sit outside this scope by design, certified combinatorially. An obstruction taxonomy (physical, epistemic, institutional) is instantiated on the audited grid, and a calibrated two-stock resource model — one fixed control set throughout — yields a closed-form critical aggregate $Y^* = \frac{27}{5}$: aggregate-yield fibres are viable exactly up to Y^* , the obstruction at $Y = 6$ is certified by a two-point dual measure with margin $\frac{3}{50}$, and the fibre-width formula $\Delta x_1(Y) = Y - 4$ gives the aggregation rule its exact reading. Dynamic-envelope, quantization, and stochastic-bridge extensions are recorded as conjectures; the residual dynamic-observation completeness gap of the source paper remains open. All claims are machine-verified in exact rational arithmetic (8 check families).

Introduction

The obstruction calculus’s certificates are emptiness statements: a belief is obstructed when a correspondence of jointly acceptable responses is empty (Abaee, 2026b, Theorem 2; Abaee, 2026c). For polyhedral safe-action sets the emptiness carries an explicit dual object — the Farkas multiplier vector — so that the obstruction itself becomes a checkable artifact; the sparse-witness bound (at most $m+1$ compatible states) and the singleton Isaacs drift complete the static picture. This paper unifies the static layer: the measure dual of the common-action obstruction,

of which the normalized Farkas pair, the sparse witness, and the Isaacs reduction are special cases. The measure dual is verified exactly; the sparsity bound is proved by a finite-reduction route (finite reduction, linear-programming duality, basic-solution support); the singleton Isaacs reading is stated with its load-bearing distributional caveat; the equality’s convexity boundary is exhibited by counterexample; the state-versus-epistemic separation is adopted as a working axiom; and a calibrated two-stock benchmark with a single fixed control set is solved in closed form. The dynamic layer — an adversarial-prior envelope for the restricted-information game — is then formulated and analyzed, and the residual dynamic-observation completeness gap (Abaee, 2026b, Open Problem 1) is *not* closed here.

Table 1: The theorem and its check families: what the verification record certifies.

Family	Statement certified	Checks
G1	general tower on all trees ($K=3$, adaptive filtration); martingale property at every cell, all 8192 policies	
G2	general DP equals path enumeration ($K=2$, 19683 policies); $Q_0 = N_1$ equals the tree supremum on both instances	
G3	refinement monotonicity, strict on a crafted functional	
G4	bridge direction: $Q_0 < 0$ defeats every policy measurably	
G5	coupling selection is real: 0 vs +1 at equal marginals	
G6	text, retired markers, proof pairing	

The measure dual of the common-action obstruction

Let the safe set be $V = \{x : q_j(x) \geq 0, j = 1, \dots, m\}$ with q_j of class C^1 ; let D be a compact-valued upper semi-continuous disturbance correspondence; let the control set $U \subseteq \mathbb{R}^k$ be compact and *convex*; and let the dynamics be affine in the control, $f(x, u, d) = f_0(x, d) + G(x)u$, with f_0 and G continuous. For a compact information set $B \subseteq V$ (a fibre or estimation tube) with nonempty active bundle

$$\mathcal{K}(B) = \{(x, j, d) : x \in B, q_j(x) = 0, d \in D(x)\},$$

define the inward drift $\psi(x, j, d; u) = \nabla q_j(x) \cdot [f_0(x, d) + G(x)u]$ and the expected drift $\Phi(u, \mu) = \int_{\mathcal{K}(B)} \psi d\mu$ over Borel probability measures μ on $\mathcal{K}(B)$.

Theorem 1 (minimax dual certificate). Under the standing hypotheses the common safe-action set $\mathcal{R}_V^B(B) = \bigcap_{x \in B} \mathcal{R}_V(x)$ is empty if and only if there exist an adversarial measure $\mu^* \in \mathcal{P}(\mathcal{K}(B))$ and a margin $\varepsilon > 0$ with

$$\max_{u \in U} \Phi(u, \mu^*) \leq -\varepsilon < 0. \quad (\star)$$

Moreover $\max_{u \in U} \min_{\mu \in \mathcal{K}(B)} \psi = \min_{\mu} \max_{u \in U} \Phi(u, \mu)$.

Proof. $\mathcal{P}(\mathcal{K}(B))$ is weak-* compact and convex (Banach – Alaoglu), and Φ is continuous, affine in u (control affinity), and linear in μ ; Sion’s minimax theorem applies, and the inner infimum over measures of a linear functional is attained by the Dirac measures, so $\max_u \min_{\mu} \Phi = \max_u \min_{\mu \in \mathcal{K}(B)} \psi = \min_{\mu} \max_u \Phi$. For the equivalence: $u \in \mathcal{R}_V^B(B)$ exactly when $\min_{\mu \in \mathcal{K}(B)} \psi(x, j, d; u) \geq 0$; the minimum is continuous in u , so nonemptiness of the common safe-action set is equivalent to $\max_{u \in U} \min_{\mu \in \mathcal{K}(B)} \psi \geq 0$ (the maximum is attained on the compact convex U), and emptiness is equivalent to the common value being strictly negative — which, by the equality, is equivalent to some μ^* with $\max_u \Phi(u, \mu^*) < 0$; the margin is the slack. $\square$

Proposition 1 (atomic support; $k + 1$ witnesses, tight). If $\mathcal{R}_V^B(B) = \emptyset$, the certificate μ^* may be chosen atomic with support of at most $k + 1$ triples, and the bound is tight in dimension k .

Proof. Write $\psi(x, j, d; u) = v(x, j, d) \cdot u + c(x, j, d)$ with $v = G(x)^\top \nabla q_j(x)$ and $c = \nabla q_j(x) \cdot f_0(x, d)$. On the compact set $U \times \mathcal{K}(B)$ the function $-\psi$ is continuous with $\min_u \max_{\mu \in \mathcal{K}(B)} (-\psi) = 2\eta > 0$ (Theorem 1); a standard finite-cover argument extracts finitely many triples (x_i, j_i, d_i) , $i \leq N$, whose drift rows still satisfy $\min_{u \in U} \max_i (-\psi_i) \geq \eta$. Min–max duality for the finite system (linear programming duality over the simplex) yields $\lambda \geq 0$, $\sum_i \lambda_i = 1$, with $\sum_i \lambda_i \psi_i(u) < 0$ on U ; an extreme such λ is a basic solution of $k + 1$ active constraints and is supported on at most $k + 1$ indices (standard theory: Chvátal, 1983, chapters 2 and 7; the pairing argument itself is Carathéodory’s theorem in $\mathbb{R}^k$). The tightness bound is Helly’s theorem in the plane (Helly, 1923) applied to the row half-spaces. Tightness: in dimension $k = 2$ the rows $\{u_1 \geq 1\}$, $\{u_2 \geq 1\}$, $\{u_1 + u_2 \leq 1\}$ on the compact square $[-10, 10]^2$ are pairwise intersecting with empty total intersection (checked on the rational grid by the record), so no two witnesses suffice; for $k = 1$ the instance of E1 certifies with exactly two atoms. $\square$

Proposition 2 (recoveries). (i) *Polyhedral*. If the drift rows are affine with rational data, the measure certificate is the ℓ_1 -normalized Farkas pair of Abaee, 2026b, Section 3.2: on the two-floor instance $\{u \leq \frac{2}{5}\} \cap \{u \geq \frac{3}{5}\}$ with $U = [0, 1]$, the measure $\lambda = (\frac{1}{2}, \frac{1}{2})$ makes the expected drift constant $-\frac{1}{10}$, the normalized multiplier pair of the worked certificate in Section 3.4 there; on the feasible contrast $\{u \leq \frac{2}{5}\} \cap \{u \geq \frac{1}{5}\}$ both sides of Theorem 1 equal $+\frac{1}{10}$ and no certificate exists. (ii) *Singleton*. For $B = \{x\}$ the dual measures live on the disturbance set; on the exact convex instance $\psi(u, d_1) = 1 - 2u$, $\psi(u, d_2) = 2u - 1$, $\max_u \min_d \psi = \min_{\mu} \max_u \Phi = 0$, while the single-worst-disturbance reading $\min_d \max_u \psi = 1$ is *not* the dual: the adversarial distribution over disturbances is load-bearing, and the Isaacs drift is its Dirac-supported inner infimum.

Proposition 3 (convexity boundary). The equivalence of Theorem 1 fails without control convexity. With $U = \{0, 1\}$ and drift rows $(-1, +1)$, $(+1, -1)$, the common safe-action set is empty, yet $\max_u \min \psi = -1$ while $\min_{\mu} \max_u \Phi = 0$: a strict minimax gap, and no adversarial measure certifies. The

audited finite action codices of the catalogue notes are non-convex by construction; there the obstructions are certified combinatorially, and the present theorem is a companion for the continuous, control-affine layer.

The obstruction taxonomy

The standing axiom adopted from the audit: *a dead state is never an information failure*. Three classes of nonviability are distinguished and, on the audited grid, are instantiated side by side (all exact): *physical* — the corner $(1, 1)$ under two floors is nonviable as a singleton under full information (no instrument keeps both floors); *epistemic* — the audited belief $\{(1, 2), (2, 1)\}$ has every branch viable and the joint set empty (failure by merge); *institutional* — under the unanimity protocol with agency 2 restricted to vote 1, the decentralized kernel collapses from 12 to 0 (failure by protocol; full enumeration). The taxonomy disciplines attribution across the catalogue: mode-blind losses on the same grid are generated entirely at the singleton level, and no count that mixes the classes is admissible as an epistemic claim.

A calibrated two-stock benchmark

Leaving the audited grid, the continuous layer needs a worked system with one fixed control set throughout. Two renewable stocks ($r_1 = 1$, $r_2 = \frac{4}{5}$, $K_1 = K_2 = 10$, cross-competition $\frac{1}{20}$, shocks $|d_i| \leq \frac{1}{10}$), floors $x_i \geq 2$, and the single control set $U = \{u \geq 0 : u_1 + u_2 \geq 2\}$ (a demand floor; no capacity constraint is imposed). On the aggregate-yield fibre $Y = x_1 + x_2$ within the floors, the active-floor harvest caps are

$$\text{cap}_1(Y) = \frac{3}{2} - \frac{Y-2}{10}, \quad \text{cap}_2(Y) = \frac{59}{50} - \frac{Y-2}{10},$$

so the fibre is viable exactly when $\text{cap}_1(Y) + \text{cap}_2(Y) \geq 2$, i.e.

Theorem 2 (critical aggregate). $Y^* = \frac{27}{5}$: aggregate fibres with $Y \leq \frac{27}{5}$ are viable and those with $Y > \frac{27}{5}$ are obstructed, the value being exact ($\text{cap}_1 + \text{cap}_2 = 2$ at Y^*). At $Y = 5$ the fibre is viable with witness $u = (\frac{6}{5}, \frac{4}{5})$; at $Y = 6$ it is obstructed ($\text{cap sum } \frac{47}{25} < 2$) with dual certificate $\mu^* = (\frac{1}{2}, \frac{1}{2})$ on the two active-floor atoms — two points, within the $k + 1 = 3$ bound — and margin $\varepsilon = \frac{3}{50}$. The fibre criterion itself — viability exactly on the cap-sum inequality, audited per aggregate with radius bounds — is carried in the worked-systems companion (Abaee, 2026c); this section contributes the two-stock derivation and the fibre-width formula.

The reading is the opposite of the intuition that abundance certifies safety: cross-competition tightens the harvest caps as the aggregate grows, so richer fibres obstruct first — aggregate yield is a failing indicator in both directions. The fibre-width formula makes the aggregation rule exact: the Y -fibre within the floors is $x_1 \in [2, Y - 2]$, of width $\Delta x_1(Y) = Y - 4$, so a linear index cannot separate states closer together than the floor buffer — the composite index is certifying exactly when the width stays inside the buffer, and the certainly-safe reporting of the applications companion (Abaee, 2026a) governs otherwise.

The dynamic envelope

A natural candidate formulation of a dynamic certificate suggests itself: an adversarial measure curve $t \mapsto \mu_t^*$ whose expected-margin envelope $Q(t)$ decreases at a certified rate, governed by an information-set operator acting on the state.

The candidate fails, for three reasons that shape the formulation developed below; this section records the reasons, formulates the envelope, and proves what exact methods reach. The residue remains conjectural.

Three obstructions to the candidate

(i) *A Bellman object posed as a pointwise Hamiltonian.* The candidate operator maps a value along trajectories to an expectation over an information set of measures — a Bellman-type, hence nonlocal, object — while dressed as a pointwise differential operator. Restricted-information games do not produce pointwise Hamiltonians in the state; their well-posed home is the information state (here, the belief), on which the object is a dynamic program. The nonlocality is not itself the obstruction; the pointwise differential form is.

(ii) *An inconsistent measure curve.* A curve $t \mapsto \mu_t^*$ of measures carries no requirement that μ_t^* be the marginal at time t of one measure on disturbance paths. Without that consistency the envelope has no time consistency: nothing connects $Q(t)$ to $Q(t')$ for $t' > t$, so no backward recursion and no comparison can even be posed. The formulation developed below replaces the curve by a single adversarial prior μ^* on disturbance paths; conditional expectation then supplies the martingale structure without any PDE.

(iii) *Expectation versus trajectory.* For the certificate direction the selection argument is not needed at all: an expectation of a bounded functional that is strictly negative exhibits a positive-measure set of realizations on which the functional is strictly negative. Selection theorems are required only for the converse value-attainment direction, which the certificate does not use.

The envelope formulation

Lemma 1 (expectation-to-realization bridge). Let μ^* be a probability measure on disturbance paths compatible with the information structure, let π be nonanticipative, and let the safety functional $F^\pi(\omega)$ — the realized weighted facet margin at the horizon — be bounded below and $\mathcal{F}_T$ -measurable. If $\mathbb{E}^{\mu^*}[F^\pi] < 0$, then $\mu^*(\{\omega : F^\pi(\omega) < 0\}) > 0$: some compatible disturbance realization violates the margin, and the set of such realizations carries positive μ^* -measure.

Proof. Nonanticipativity of π makes F^π an $\mathcal{F}_T$ -measurable functional of the disturbance path, and boundedness below makes its expectation well-defined. If $\mu^*(F^\pi < 0) = 0$, then $F^\pi \geq 0$ holds μ^* -almost surely, so $\mathbb{E}^{\mu^*}[F^\pi] \geq 0$ — contradicting the hypothesis. The realizations of the positive-measure set are compatible by construction, since μ^* is supported on the compatible set. $\square$

Definition 1 (adversarial-prior envelope). Fix a horizon T , a bounded safety functional F (positive values safe), and one consistent adversarial prior μ^* on disturbance paths. The envelope is

$$Q(t) = \operatorname{ess\,sup}_{\pi} \mathbb{E}^{\mu^*}[F^\pi \mid \mathcal{F}_t],$$

the essential supremum over nonanticipative policies of the conditional expected safety functional. For each fixed π the process $t \mapsto \mathbb{E}^{\mu^*}[F^\pi \mid \mathcal{F}_t]$ is a martingale; the envelope is its essential supremum over policies, and its evaluation at information states — a Bellman recursion, not a differential operator — is the well-posed successor of the proposed information-set operator.

Theorem 3 (finite well-posedness, comparison, exactness). In the finite setting — disturbance set Ω finite, information states

b drawn from a finite partition, admissible actions finite, data rational — define the one-step operator

$$\Phi V(b) = \max_a \mathbb{E}^{\mu^*}[V(b') \mid b, a],$$

where b' is the posterior given (b, a, y) the maximum over admissible actions of the prior-averaged next-belief value. Then: (i) Φ is order-preserving, $V \leq W \Rightarrow \Phi V \leq \Phi W$ pointwise; (ii) $Q(0) = \Phi^N g$ is computed exactly by backward induction, in rational arithmetic; and (iii) $Q(0) < 0$ certifies that for every admissible policy the set of disturbance realizations defeating it carries positive μ^* -measure.

Proof. (i) If $V \leq W$ pointwise, then for each (b, a) the conditional expectation of $V(b')$ is at most that of $W(b')$ (monotonicity of expectation over the same kernel), and the maximum over the same action set preserves the inequality. (ii) By induction on the remaining horizon: $\Phi^0 g = g$ is exact by definition; the induction step applies Φ once, and each evaluation is a finite weighted sum of previously computed values, so rational data propagate exactly through the recursion. Well-posedness — existence and uniqueness of the recursion's output — is supplied by the same induction. (iii) $Q(0) < 0$ bounds every policy's conditional expectation: for the optimal policy the expectation is $Q(0) < 0$, and for every suboptimal π the expectation is smaller still, so Lemma 1 applies to each π and yields the positive-measure defeating set. $\square$

Proposition 4 (identification with the recourse certificate). One step deep, with stored facet margins β_ℓ , control kernels $k_\ell(\sigma)$, prior λ over the stored branches, and control set U , the envelope evaluation of Definition 1 — the policy's optimal ceiling on the expected weighted facet sum — is exactly the support-function certificate

$$\Gamma_h(\lambda) = \sum_{\ell} \lambda_{\ell} \beta_{\ell} + \int_0^{\tau} h_U \left(\sum_{\ell} \lambda_{\ell} k_{\ell} \right) + \sum_{\ell} \int_{\tau}^T h_U(\lambda_{\ell} k_{\ell}),$$

with $h_U(g) = \max_{v \in U} g^{\top} v$ the support function of U , of the obstruction calculus (its Section 3.7; complete proof in its Supplementary Material, S1). The recourse certificate is the dynamic envelope's one-step evaluation.

Proof. Variation of constants expands each realization of the weighted facet sum into the stored β -term plus the window pairing (one common signal on $[0, \tau)$) plus the branch pairings (separate post-reveal signals). The supremum of a linear functional over the signal sets factorizes across the time intervals, and each interval's supremum of a realized pairing $g^{\top} v$ over $v \in U$ is the support value $h_U(g)$; collecting terms gives exactly $\Gamma_h(\lambda)$ as the ceiling of every policy's realized weighted facet sum. Negativity of the ceiling certifies by Lemma 1: the realized weighted facet sum of every policy is at most $\Gamma_h(\lambda) < 0$, so some facet is violated on a positive-measure set of realizations. $\square$

Corollary 1 (the three-branch instance, certified). On the braking instance of the obstruction calculus (three branches, weights $\lambda = (\frac{3}{8}, \frac{5}{16}, \frac{5}{16})$, stored margin $\beta = \frac{16}{25} - \tau - 1$): the pooled window kernel vanishes ($\sum_{\ell} \lambda_{\ell} n_{\ell} = 0$), each branch's post-reveal pairing contributes its support value $\lambda_{\ell} \int_{\tau}^{\tau+1} (\tau + 1 - \sigma) d\sigma = \lambda_{\ell}/2$, and the envelope evaluation is $\Gamma_h(\tau) = \frac{7}{50} - \tau$: negative exactly for $\tau > \frac{7}{50} = 0.14$, where the bridge converts it into the positive-measure defeating set for every blind-window policy; hold-then-brake is safe exactly to $\tau = \frac{7}{50}$, so the certificate is tight. The verification script evaluates the identity exactly at $\tau \in \{\frac{1}{10}, \frac{7}{50}, \frac{1}{5}\}$.

Conjecture 1 (the open residue in continuous time). The continuous-time residue of the envelope formulation: (i) an evolution law for the adversarial prior along the review sequence

— a consistent coupling of the one-step priors, in the spirit of martingale-transport couplings — under which the envelope obeys a dynamic-programming equation in continuous time; (ii) the limit of the information-state recursion under partition refinement, which is where a comparison principle must live if a differential statement is to be recovered at all; and (iii) value attainment (the selection direction) in the general filtered setting. The static identification of Proposition 4 is exact and is not part of the conjecture.

The review-sequence envelope: exact couplings

The envelope of the preceding section is one review deep. This section certifies the two properties its continuous-time residue must preserve — coupling consistency and refinement monotonicity — on exact review-sequence instances; everything is rational arithmetic, regenerated by the script.

The instance

Two windows, one hidden branch. The branch is $s \in \{+1, -1\}$, prior $\frac{1}{2}$ each; the state starts at $z_0 = 2$. At each review the policy acts $u_1, u_2 \in \{-2, 0, 2\}$; between reviews the adversarial disturbance acts $d_1, d_2 \in \{-1, 1\}$ (one consistent prior: the product measure, $\frac{1}{2}$ per value per window). The state obeys $z_1 = z_0 + s u_1 + d_1$ and $z_2 = z_1 + s u_2 + d_2$; the review after window 1 observes z_1 exactly. The safety functional is the expected two-sided margin

$$F = \mathbb{E}_s[2 - |z_2(s)|], \quad z_2(s) = z_0 + s(u_1 + u_2) + d_1 + d_2,$$

bounded on the finite instance, so the bridge lemma applies verbatim.

Proposition 5 (consistent coupling and the tower identity). The one-step adversarial priors couple to the product measure on (d_1, d_2) , and for *every* nonanticipative policy on the instance — all $3^4 = 81$ policy trees over the reachable z_1 cells — the path-enumerated expectation and the iterated-conditional (tower) evaluation of $\mathbb{E}[F]$ agree exactly. Each fixed policy's conditional-expectation process along the review filtration is therefore a martingale, and the envelope $Q_0 = \text{ess sup}_\pi \mathbb{E}[F \mid \mathcal{F}_0]$ is computed by backward induction: $Q_0 = \Phi^2 g$ equals the policy-tree supremum ($= \frac{1}{2}$, at $u_1 = \pm 2$).

Proof. Consistency is by construction: the product measure has the one-step priors as its marginals. For the tower identity, condition on $\mathcal{F}_1$ (the review): within each z_1 cell the inner conditional expectation is a finite weighted average, and the outer average over the cells reconstructs the path sum because the cells partition the paths — an induction on the two-node filtration, exact in rationals. The martingale property is the tower identity restated along the filtration. Backward induction computes $\Phi^2 g$: Φg at each cell is a maximum of three rational expectations, and Φ is applied once more over u_1 ; the script checks the result against the enumeration of all 81 trees. $\square$

Proposition 6 (information design and recourse strictly pay). On the instance: the envelope value is $Q_0 = \frac{1}{2}$, attained by $u_1 = -2$ (any $u_1 = \pm 2$ by symmetry) with the recourse $u_2(z_1) = (0, -2, +2, +2)$ at $z_1 = (-1, 1, 3, 5)$; every open-loop policy and the do-nothing policy attain only 0. The strict gap $\frac{1}{2} > 0$ is the certified content of the correction: the first review's action *designs the information* ($u_1 = -2$ makes z_1 separate the branch exactly), and the second *exploits it* (brake toward the origin on the revealed branch); at $z_1 = 5$ the revealed state is unrecoverable and the cell value is -1 .

Proof. The four singleton cells at $u_1 = -2$ carry values $Q_1 = (1, 1, 1, -1)$ at weights $\frac{1}{4}$ each: $Q_0 = \frac{1}{2}$. The open-loop maximum is the better of nine fixed pairs, computed exactly as 0; the do-nothing pair $(0, 0)$ also evaluates to 0. The cell values and the maximizing u_2 are direct rational computations, reproduced by the script. $\square$

Proposition 7 (refinement is monotone). On the three-state matching instance — hidden $x \in \{1, 2, 3\}$, prior $\frac{1}{3}$ each, margin $-|x - a|$, action a per observation block — the envelope is non-decreasing under partition refinement and strictly increasing on the instance's final step: $Q(\{\{1, 2\}, \{3\}\}) = Q(\{\{1\}, \{2, 3\}\}) = -\frac{1}{3} < 0 = Q(\{\{1\}, \{2\}, \{3\}\})$.

Proof. Every policy measurable under a coarser partition is measurable under a refinement, so the supima are ordered. Strictness: under the fine partition the policy matches each state exactly ($a = x$, value 0); under either coarse partition two states share an action and the best shared action leaves one mismatch, value $-\frac{1}{3}$. This monotone, bounded family is the direction the partition-refinement limit of the residue conjecture must take. $\square$

The residue conjecture now reads sharper: the continuous-time coupling law must reduce, on review sequences with finite disturbance sets, to the product coupling certified here; and the information-state recursion must converge along refinements with the monotonicity certified here.

The general finite-sequence envelope

The instance of the previous section is the $K = 2$ case of a general statement. This section states the general theorem over finite review sequences and proves it in full; the two-window instance is the $K = 2$ case, and the general argument uses only finite combinatorics and the linearity of expectation.

Setup

Fix a number of windows $K \geq 1$. Each window k carries a finite disturbance set D_k ; the disturbance paths form $\Omega = D_1 \times \cdots \times D_K$, a finite set. A *consistent adversarial prior* is a product measure $\mu^* = \mu_1^* \otimes \cdots \otimes \mu_K^*$ on Ω ; consistency — each marginal μ_k^* is the time- k law of the one measure μ^* — is exactly the requirement whose absence makes a backward recursion ill-posed. Reviews divide the sequence: the review filtration is an increasing sequence of partitions $\mathcal{P}_1 \preceq \mathcal{P}_2 \preceq \cdots \preceq \mathcal{P}_K$ of Ω , each $\mathcal{P}_k$ consisting of the cells of the information available at review k (so $\mathcal{P}_k$ refines into $\mathcal{P}_{k+1}$), with $\mathcal{P}_K$ the singleton partition; $\mathcal{P}_1$ may be the trivial partition (a blind first window). Window k has a finite action set A_k ; a *nonanticipative policy* is $\pi = (\pi_1, \dots, \pi_K)$ with π_k constant on the cells of $\mathcal{P}_k$ — the filtration carries the whole observation history, so cell-constancy is adaptedness. The data are completed by a bounded safety functional $G : \Omega \times A_1 \times \cdots \times A_K \rightarrow \mathbb{R}$ (positive values safe, per the sign convention of this paper), rational throughout; the realized functional of π is $G^\pi(\omega) = G(\omega, \pi_1[\omega]_1, \dots, \pi_K[\omega]_K)$, where $[\omega]_k$ is the $\mathcal{P}_k$ -cell of ω .

Theorem 4 (general finite-sequence envelope). Under the setup above: (i) *tower and martingale*. For every nonanticipative policy π ,

$$\mathbb{E}^{\mu^*}[G^\pi] = \mathbb{E}^{\mu^*}[\mathbb{E}^{\mu^*}[\cdots \mathbb{E}^{\mu^*}[G^\pi \mid \mathcal{P}_K] \cdots \mid \mathcal{P}_2] \mid \mathcal{P}_1],$$

and $M_k^\pi = \mathbb{E}^{\mu^*}[G^\pi \mid \mathcal{P}_k]$ is a μ^* -martingale along the filtration. (ii) *Dynamic programming*. Define the node values N on the review tree — nodes are pairs (cell, action prefix) — by

$N_K(\{\omega\}, a_{1:K-1}) = \max_{a \in A_K} G(\omega, a_{1:K-1}, a)$ and, for $k < K$, the sum running over the children $C' \subseteq C$ of $\mathcal{P}_{k+1}$ with the child weights $w(C') = \mu^*(C')/\mu^*(C)$,

$$N_k(C, a_{1:k-1}) = \max_{a \in A_k} \sum_{C'} w(C') N_{k+1}(C', a_{1:k-1}a),$$

Then the envelope value $Q_0 := \sup_{\pi} \mathbb{E}^{\mu^*}[G^{\pi}]$ equals N_1 evaluated on the $\mathcal{P}_1$ -cells, averaged by their prior weights $Q_0 = \sum_{C \in \mathcal{P}_1} \mu^*(C) N_1(C)$ (a single node if $\mathcal{P}_1$ is trivial); the supremum is attained by a nonanticipative policy read off the maximizing arguments of N . (iii) *Refinement monotonicity*. If a second filtration $\mathcal{P}'$ refines $\mathcal{P}$ stage-wise, then the induced envelope values satisfy $Q_0(\mathcal{P}) \leq Q_0(\mathcal{P}')$. (iv) *Exactness and the obstruction direction*. All node values are rational; if $Q_0 < 0$, then for every nonanticipative π the set $\{\omega : G^{\pi}(\omega) < 0\}$ carries positive μ^* -measure.

Proof. (i) Each G^{π} is $\mathcal{P}_K$ -measurable (a finite function of the path and the cell-constant actions), and repeated conditioning is an identity on finite probability spaces; the tower formula is that identity read backwards, and the martingale property is $M_k^{\pi} = \mathbb{E}[M_{k+1}^{\pi} \mid \mathcal{P}_k]$, which is the defining property of conditional expectation applied twice. (ii) By backward induction on the stage. For $k = K$ the node value is the best terminal action, and G^{π} at a singleton cell is $G(\omega, \cdot)$ evaluated at the realized prefix and π_K , so $\max_{a \in A_K} G = N_K$ pointwise. For the step, fix a $\mathcal{P}_k$ -cell C and an action prefix, and split the policies by their stage- k action: every policy compatible with the prefix is a choice $a \in A_k$ together with a continuation policy for the conditioned sub-instance on $\{C\}$ -successor cells, whose filtration $(\mathcal{P}_j)_{j>k}$ restricted to the successors refines to singletons at K . By (i) applied at the conditioning,

$$\mathbb{E}^{\mu^*}[G^{\pi} \mid C] = \sum_{C' \subseteq C} \frac{\mu^*(C')}{\mu^*(C)} \mathbb{E}^{\mu^*}[G^{\pi} \mid C'],$$

the conditional law of the successor cell given C is the ratio of prior masses (exogeneity: the prefix does not move μ^*), and the induction hypothesis — applied to each conditioned sub-instance, which has one fewer window — identifies the supremum of the inner conditional expectations over continuations of $(a_{1:k-1}, a)$ with $N_{k+1}(C', a_{1:k-1}a)$. Taking the outer maximum over a gives N_k ; summing over the $\mathcal{P}_1$ -cells with their prior weights and applying (i) once more gives $Q_0 = \sum_C \mu^*(C) N_1(C)$. Attainment: each N node is a maximum over a finite set; selecting a maximizing argument at every node and pasting along the tree yields a nonanticipative policy — cell-constancy holds because the selection is made per node — and its expectation is exactly the recursion’s value. (iii) If $\mathcal{P}'$ refines $\mathcal{P}$ stage-wise, every $\mathcal{P}$ -cell-constant policy is $\mathcal{P}'$ -cell-constant, so the $\mathcal{P}'$ -supremum ranges over a superset of policies and cannot be smaller; the inequality for the full sequences follows stage-wise from the one-step inclusion. (iv) The recursion’s only operations are maxima and μ^* -weighted averages of rationals, so every N is rational. If $Q_0 < 0$, then by (ii) every policy satisfies $\mathbb{E}^{\mu^*}[G^{\pi}] \leq Q_0 < 0$, and the expectation-to-realization bridge (Lemma 1) converts strict negative expectation into a positive-measure set of strictly negative realizations. $\square$

Remark 1 (the recourse certificate and the residue). Three consequences. *First*, the identification with the recourse certificate is the one-step case of Theorem 4: with stored facet margins the node recursion collapses to the support-function evaluation Γ_h (the identification proposition), so the calculus’s recourse certificate is the envelope’s K -node tree read at depth one. *Second*, the one-step priors μ_k^* do not determine the coupling μ^* : two consistent couplings with identical one-step marginals can give different node values (the script

certifies a two-window example where the product coupling yields 0 and the comonotone coupling yields +1 on the same functional). The residue conjecture’s “evolution law for the adversarial prior” is therefore precisely a *coupling-selection* problem — choose the consistent coupling as a function of the state and review history — which is the finite-space form of the martingale-transport structure recorded in the conjecture. *Third*, refinement monotonicity (iii) is the finite-space grounding of the partition-limit clause: the instance family of the previous section, monotone and bounded, now carries a general ordering, and the continuous-time residue is exactly the question of which couplings and which refinement limits survive the passage to continuous disturbance sets. The distributionally robust interpolation of the probabilistic companion (Abae, 2026e) holds the adversary to a mixture family: on the two-window instance its value is exactly

ρ at parameter

ρ — the interpolation contains the coupling extremum at $\rho = 1$, preserves the one-step marginals at every ρ , and leaves the coupling-selection difficulty orthogonal to the mixture parameter (joint check archived).

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Declarations

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